Problem 5

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A: For all real numbers x, there exists an integer n such that $n \leq x < n+1$.
B: There exists an integer n such that, for all real numbers x, $n \leq x < n+1$.

Part (a)

$$A\text{: } \forall x \in \mathbb{R}, \exists n \in \mathbb{Z} : n \leq x < n+1.$$
$$B\text{: } \exists n \in \mathbb{Z} : n \leq x < n+1 \ \forall x \in \mathbb{R}.$$

Part (b)

A is saying that all real numbers are bounded between some two consecutive integers. This statement is true. B is saying that there are consecutive integers between which all real numbers lie. This statement is not true.

Part (c)

Suppose every person has exactly one soulmate, and that everyone is the soulmate of their soulmate. Consider then these two statements:

A: For every person x, there exists another person y such that x and y are soulmates.
B: There exists some person y such that, for every person x, x and y are soulmates.

Statement A is true. It reflects the (unequivocally) true property that everyone has a soulmate. However, statement B suggests that there is some person that is *e*veryone's soulmate. But every person (undoubtedly) only has one soulmate, so this is false.